STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 10C

Description

10 c) Write R program to find Decision Tree with the sample data and visualize the regression graphically.

Procedure and Overvation:

20R01A6753

Description:

Decision tree is a graph to represent choices and their results in form of a tree. the nodes in the graph

represent an event or choice and the edges of the graph represent the decision rules or conditions.

It is mostly used in machine learning and data mining appliations using R.

EXAMPLE:

Use of decision trees is predicting an email as spam or not spam.Predicting of a Tumor is cancerous or predicting o f

a loan as good or bad.Credit risks based on the factors in each of these.

Generally a model is created with observed data also called training data.Then a set of validation data is used

to create and visualize decision trees.

The R package "party" is used to create decision tree.

Install R package-use the below commans in R console to install package.You also have to install the dependent pack ages if any.

Install.package("party")

The package party as the function tree() which is used to create and analyze decision tree.

Syntax:ctree(formula,data)

Where

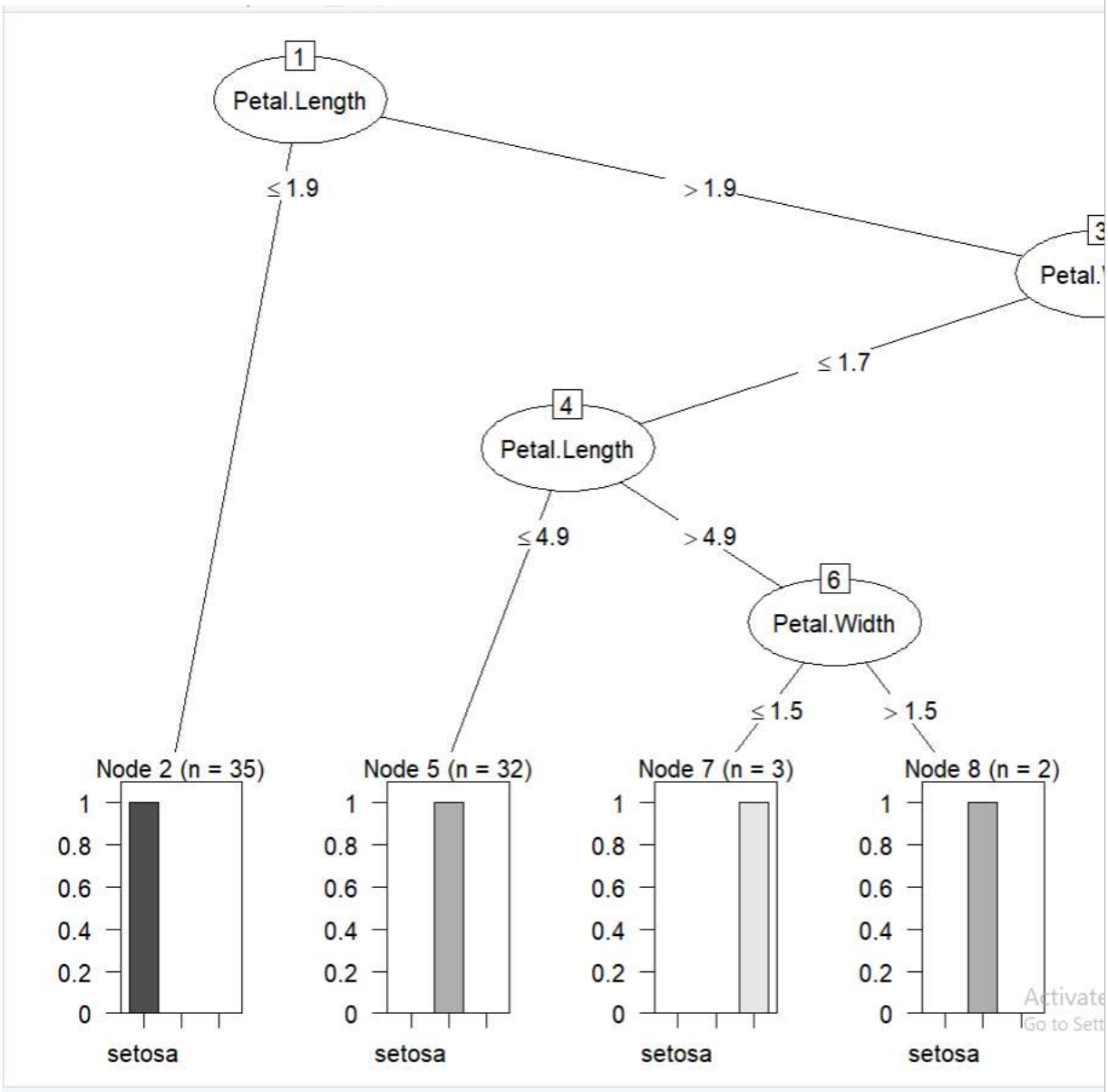
formula is describes the predictor and response variable.

data is the name of the dataset.

SOURCE CODE:

```
#Data Load

data("iris")

#Install the required packages

#install.packages("party")

install.packages("caret")

install.packages("C50")

#Library invoke

#library(party)

library(caret)

library(C50)

#To make the results consistent across the runs

set.seed(7)

#Data Partition

inTrainingloca<-createDataPartition(iris$Species,p=0.70,list=F)

training<-iris[inTrainingloca,]

testing<-iris[-inTrainingloca,]

#Model Building

model<-C5.0(Species~.,data=training)

#Generate the model summary

summary(model)
```

```
#Predict for test data set

pred<-predict.C5.0(model,testing[,-5]) #type ="prob"

#Accuracy of the algorithm

a<-table(testing$Species,pred)

sum(diag(a))/sum(a)

#Visualize the decision tree

plot(model)
```



VIVA RESPONSES

Q) ___ are used when we want to visually examine the relationship between two quantitative variables.

Scatterplot

Q) A graph that uses vertical bars to represent data is called a ____.

Bar graph

Q) Data Analysis is a process of,

All of the mentioned above

Q) Least Square Method uses ___.

Linear regression

Q) What is a hypothesis?

A statistical method for calculating the extent to which the results could have happened by chance